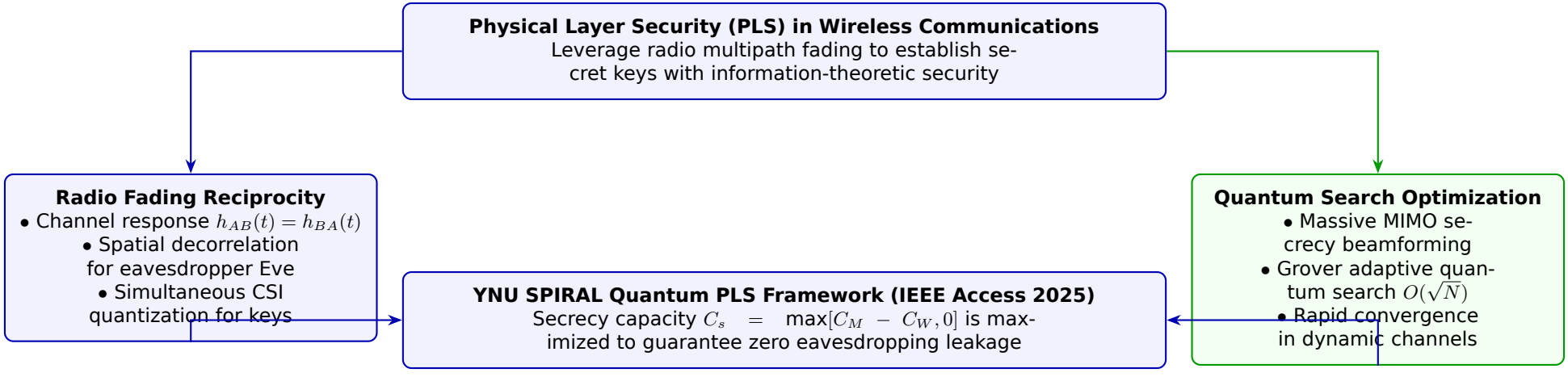

Physical Layer Security (PLS) in Wireless Communications
Leverage radio multipath fading to establish secret keys with information-theoretic security
Radio Fading Reciprocity
• Channel response $h_{AB}(t) = h_{BA}(t)$
• Spatial decorrelation for eavesdropper Eve
• Simultaneous CSI quantization for keys
YNU SPIRAL Quantum PLS Framework (IEEE Access 2025)
Secrecy capacity $C_s = \max[C_M - C_W, 0]$ is maximized to guarantee zero eavesdropping leakage
Quantum Search Optimization
• Massive MIMO secrecy beamforming
• Grover adaptive quantum search $O(\sqrt{N})$
• Rapid convergence in dynamic channels